25900256 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBOT202 - THALLOPHYTES AND ARCHEGONIATES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **1** marks

1. Name the pigment found in Sargassum. [K] / [CO1]
2. What is the specialized spore bearing structure in members of pterophyta? [U] / [CO3]
3. Name the spore-producing structures in Pteris. [U] / [CO3]
4. Which pigment gives Cladophora its green colour? [A] / [CO3]
5. Distinguish between the beneficial and harmful economic importance of fungi [E] / [CO5]
6. What is the shape of chloroplast in *Volvox*? [K] / [CO1]
7. What is the role of fungal partner in Lichen? [A] / [CO1, CO3, CO4, CO5]
8. Which bryophyte is commercially important for peat production? [K] / [CO5]
9. Write one example of edible algae. [K] / [CO5]

10. Give name of a lichen which is used in beverage production. [U] / [CO5]

11. Name a heterotrophic thallophyte. [K] / [CO2]

12. Name the asexual spores produced by fungi. [K] / [CO3]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Describe the distinguishing features of Rhodophyceae according to Fritsch. [An] / [CO3]

14. Explain the importance of Archegoniates in plant evolution. [An] / [CO2]

15. Give an account of three subclasses of class Bryopsida as per Rothmaler's classification. [U] / [CO3]

16. Explain the morphology of Polysiphonia. [U] / [CO1, CO3, CO5]

17. 'Pteridophytes are both ecologically vital and economically useful.' Evaluate this statement with examples. [E] / [CO1]

18. Explain the vegetative features of gymnosperms with suitable examples. [K] / [CO4]

19. Distinguish between vegetative cells and reproductive cells in Oedogonium. [U] / [CO1]

20. Write any five distinguishing characters of Puccinia. [K] / [CO3]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Analyze the significance of antheridia and archegonia in Riccia [An] / [CO3]

22. Elaborate on the life cycle of Xylaria. [U] / [CO4]

[10x1 = 10]